

ogies that are generally cheaper and simpler to use. Initially, in terms of performance, these disruptive technologies are only suitable for a small number of users. Over time, these technologies improve and become a necessity for most people.

This phenomenon has occurred over time and again. One such example is of the smartphone, which is making cameras, audio players, PCs, handheld GPS, and many other things obsolete.

According to many industry stalwarts, IoT will disrupt multiple industries in the coming decades.

To predict the growth of IoT we may look at the evolution of smartphones. The timeline of phones is shown in Fig. 1.

As can be seen, there was massive advancement with each successive decade. Alongside the features in the phone's hardware, the networking technology kept improving. From 1G in the 80s, which could transfer less than 10kbps, the growth in data transfer rate kept on improving by a factor of ten.

The inception of digital cellular technology in 1990s changed the scope of mobile phones. From just a device to make calls the mobile device had added functionalities, such as gaming and GPS, and in terms of inference we got a new feature of predictive text.

Then the next big development happened in the 2000s, when the data transfer rate jumped from 64kbps to over 1Mbps. The phones now had cameras, touchscreens, accelerometers, etc. This is when mobile phone technology really disrupted the market. The processor in smartphones began catching up with the basic PCs, the phone cameras affected the sale of low-level point-and-shoot cameras, and many different gadgets were affected due to these powerful smartphones.

Smartphones were no longer merely a tool for communication but became much more than that. We

|                                         | 1980-1990                 | 1990-2000                 | 2000-2010                          | 2010-2020                    | 2021+           |
|-----------------------------------------|---------------------------|---------------------------|------------------------------------|------------------------------|-----------------|
| Communicate                             | 1G (Analog Cellular)      | 2G (Digital Cellular)     | 3G (Mobile Broadband) BT NFC Wi-Fi | 4G (Native IP Networks)      | 5G              |
| Communicate-Data                        | ~1Mbps or less            | ~10Mbps or less           | ~100Mbps or less                   | ~1GBps or less               | ~10GBps or less |
| Compute                                 | Single core ~1Ghz or less | Single core ~2Ghz or less | 2 and 4 cores ~2Ghz+               | 4 and 8 cores ~3Ghz          | 8 cores+ ~3Ghz+ |
| Sense                                   |                           |                           | Camera Touch ALS/PS Accelerometers | Fingerprint                  |                 |
| Gaming                                  |                           | Gaming                    |                                    |                              |                 |
| Location                                |                           | GPS                       |                                    |                              |                 |
| Inference                               |                           | Predictive text           |                                    | Facial and voice recognition | AR/VR           |
| Secure                                  |                           |                           | Arm TZ                             |                              |                 |
| Software                                |                           | Symbian OS                | Full web browsing, app store       |                              |                 |
| Process nodes - (microns) - Moore's Law | 3 or better               | 0.8 or better             | 0.1 or better                      | 0.02 or better               | 0.01 or better  |

Fig. 1: Mobile phone evolution: Predictor for IoT evolution

saw a very high level of integration of the various functions, including computing prowess, sense, and gaming—all of these started to disrupt several industries.

The advancement made in the first decade caused major adoption of smartphones, thus bringing a flood of manufacturers driving the cost lower and lower with each passing year. The rate of refinement and growth continued in the 2010s. The inception of 4G improved the internet speed and the processing power of mobiles was now comparable to baseline PCs.

The phone cameras started to outperform the basic cameras. And similar was the fate of music players, such as MP3 players and iPods. Smartphones were no more a novelty but a necessity.

We have just set foot in the 2020s, but by now smartphones pack enough power to compete with decent PCs, cameras, etc. Some of the latest models have higher RAM and processing power than most entry-level laptops. They offer much higher resolution than most cameras, and the 5G internet is capable of matching or even exceeding the data transfer rate of cabled internet available in many areas.

Smartphones are now secure

enough to do banking transactions and, in many countries, are being used as the primary mode for everyday transactions. However, it is too early to discuss the advancements that are likely to be made by smartphones in this decade!

## The future of IoT

IoT technology is in its second generation in terms of development. Compared to mobile phones, IoT is where the phones were in 1990s as feature phones. The IoT technology is being adopted by a few, but it is yet to become a necessity for most.

From a process node standpoint, in terms of our inferencing capability, in terms of our compute capability, we are nowhere close to gigahertz, right? We have the sub-gigahertz capability today. So, these are very early days for IoT. Very likely, as the Moore's Law progresses, the sense of integration will also progress, and there will be dislocations that vendors and players will benefit from.

We normally think of Moore's Law as digital integration, and we can say that the IoT is at a beginning of Moore's Law journey. It could possibly have a high level of function integration driven by Moore's Law. When we talk about sensors, there is a variety of analogue front-ends on